

WIDS GLOBALTHON · WILDFIRE RISK

Better features. *Better model.*

THE RECIPE

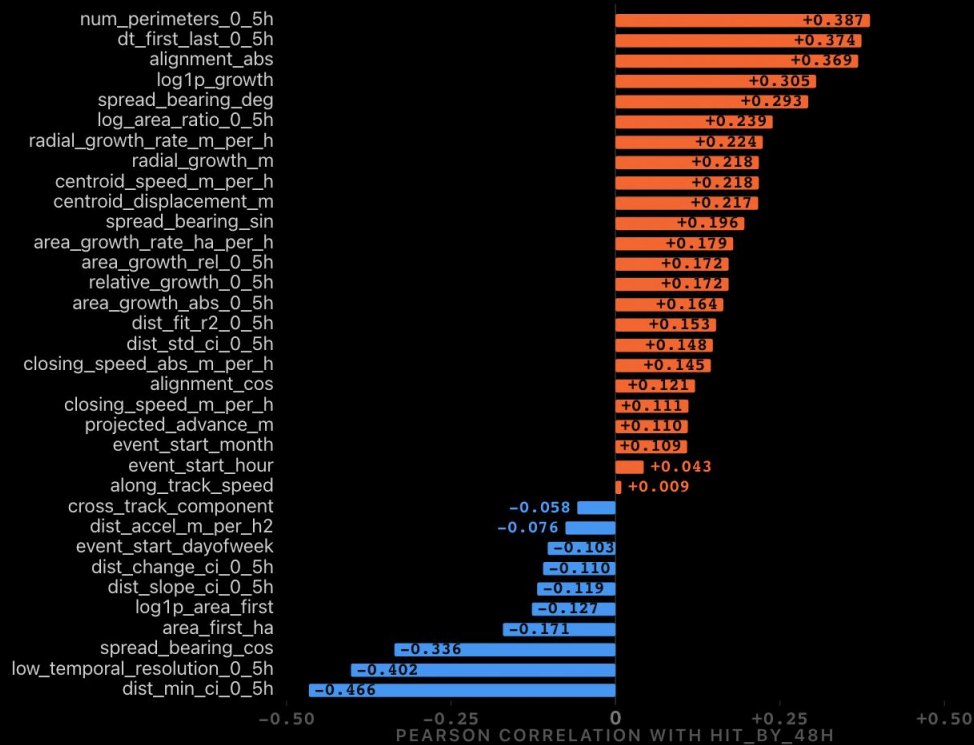
Screen. Decorrelate. Engineer. Validate.



STEP 1 SCREEN

First, see what speaks to the target.

All 34 features by Pearson correlation with `hit_by_48h`. Orange = positive (more likely to hit) · Blue = negative (less likely). A linear screen — low correlation isn't a verdict, since trees can still find non-linear signal.



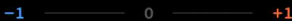
STEP 2A FIND DUPLICATES

Some signals are the *same* signal.

Across the 34-feature matrix, 25 pairs exceed $|r| > 0.85$.

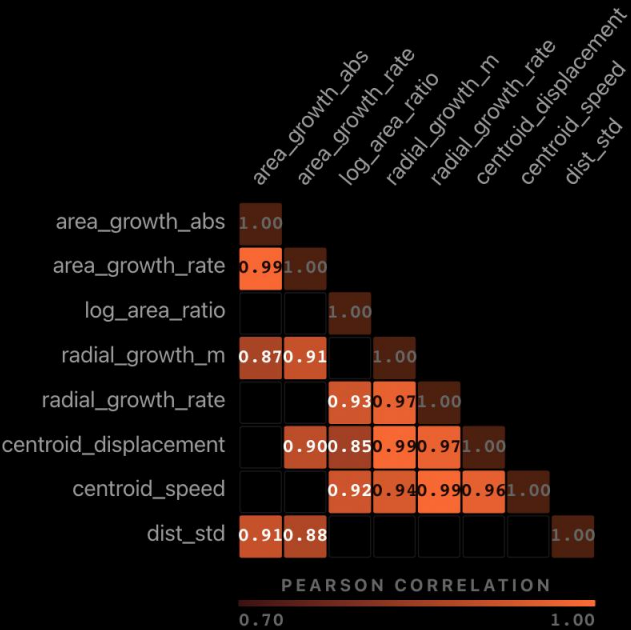
PEARSON CORRELATION COEFFICIENT

$$r = \frac{\sum_i (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_i (x_i - \bar{x})^2 \sum_i (y_i - \bar{y})^2}} \equiv \frac{\text{cov}(X, Y)}{\sigma_X \sigma_Y}$$



PERFECT INVERSE · NO LINEAR · PERFECT DIRECT

$|r| > 0.85 \Rightarrow$ two features carry essentially the same information



OUTSIDE THIS CLUSTER

1.000

relative_growth_0_5h
=
area_growth_rel_0_5h

TWO FEATURES. ONE SIGNAL.

STEP 2B DECORRELATE

Why these five are dropped.

Each row: the dropped feature, its near-perfect twin (or "near-constant" if it had none). The model needs one feature per redundant pair, not both. 34 → 29.

DROPPED		REASON	KEPT TWIN
01	closing_speed_m_per_h	$r = -0.998$ REDUNDANT	→ dist_change_ci_0_5h
02	closing_speed_abs_m_per_h	$r = +0.907$ REDUNDANT	→ area_growth_abs_0_5h
03	dist_std_ci_0_5h	$r = -0.943$ REDUNDANT	→ dist_slope_ci_0_5h
04	projected_advance_m	$r = +0.855$ REDUNDANT + SPARSE	→ area_growth_abs_0_5h
05	dist_fit_r2_0_5h	near-constant NEAR-CONSTANT	—
PER-HORIZON BRIER IMPACT · base_full → after manual cut			
12h -0.034 HELPED		24h +0.014 HURT	48h -0.001 HELPED
			72h -0.001 HELPED

5 features removed · 4 had a near-perfect twin · 1 was nearly constant 34 → 29

STEP 3 ENGINEER

Then build what raw features can't see.

The raw feature `dist_min_ci_0_5h` is the minimum fire-to-target distance observed in the first 5 hours — by far the strongest single signal in the data. Two examples below show how transforming it lifts the model's importance score even further. `29 → 36`.



STEP 4 · VALIDATE

Which feature set should we ship?

Three candidate sets — raw 34, after the manual cut (29), after engineering (36) — scored on CatBoost validation Brier (lower is better). The engineered set wins decisively on 48h and 72h, where Brier drops 4–5x vs. the raw set.

	HORIZON 12h	HORIZON 24h	HORIZON 48h	HORIZON 72h
34 · RAW ALL FEATURES	0.123	BEST 0.047	0.010	0.005
29 · + MANUAL CUT -5 REDUNDANT (SLIDE 5)	BEST 0.089	0.061	0.009	0.004
36 · + ENGINEERED +7 TRANSFORMS (SLIDE 6)	0.097	0.059	BEST 0.002	BEST 0.001

CatBoost validation Brier · lower is better

→ SHIP THE 36-FEATURE SET

THREE RULES

- 01 Cut what's *redundant*.
- 02 Build what's *missing*.
- 03 Trust *validation*.

BETTER FEATURES. BETTER MODEL.

34 → 29 → 36 · AUC 0.9987 at 48h